

Break-Aware Weighted Local Linear Forecasting for Cocoa Returns: Nonparametrics vs Machine Learning

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Abstract

This paper proposes a break-aware weighted local linear (WLL) forecasting framework for cocoa futures returns, combining nonparametric regression with structural break detection and machine-learning benchmarks. We document the empirical performance of WLL relative to random forests and gradient-boosted trees, and we analyze a rare but severe instability of WLL arising from break-aware tuning. We then show how simple regularization and tuning constraints restore the stability predicted by nonparametric theory.[0.5ex] **Keywords:** nonparametric forecasting, structural breaks, local polynomial regression, random forests, gradient boosting.

1 Introduction

2 Data and Empirical Setting

Cocoa futures provide an ideal testing ground for break-aware forecasting methods due to their unique market characteristics. As an agricultural commodity heavily concentrated in West Africa, cocoa supply is highly susceptible to climate shocks—such as El Niño events—and supply-chain disruptions. These exogenous shocks induce discrete structural breaks in the conditional mean and volatility of prices, creating a nonstationary environment that aligns well with the time-varying regression framework. The economic interpretability of these breaks, driven by known weather patterns and market events, allows us to validate statistical break detection against historical ground truth.

2.1 Cocoa returns and sample construction

We work with a univariate time series of cocoa futures prices and construct log-returns as the target variable for forecasting. Let p_t denote the settlement price of the nearby cocoa futures contract on business day t , expressed in the currency units quoted by the exchange. The log-return is defined as

$$r_t = \log p_t - \log p_{t-1}, \quad t = 2, \dots, T, \quad (1)$$

and we take $\{r_t\}_{t=2}^T$ as the basic series of interest.

The raw price data are obtained from *[DATA SOURCE, e.g. exchange or vendor]* for the period from *[START DATE]* to *[END DATE]*.¹ We construct a continuous nearby series by rolling from one contract to the next according to *[ROLL RULE, e.g. volume-based or fixed days-to-maturity]*, and we drop days on which either prices are missing or the contract does not trade. Non-trading calendar days are left as gaps rather than interpolated; all models are estimated on the resulting sequence of business days.

Throughout this paper we focus on a parsimonious univariate specification in which the predictor is a function of past returns. Formally, we treat $(x_t, y_t) = (x_t, r_t)$ with $y_t = r_t$ and x_t constructed from lagged returns (for example, $x_t = r_{t-1}$ or a small vector of lags). In the empirical implementation reported below, we adopt *[SPECIFY THE FINAL CHOICE, e.g. $x_t = r_{t-1}$]* as the baseline predictor; extensions to richer covariate sets (such as weather variables or macro indicators) are left for future work.²

2.2 Sample splitting and forecasting design

Let $\{r_t\}_{t=1}^T$ denote the full sample of returns after cleaning and continuity adjustments. We partition the sample into three conceptually distinct parts:

- (i) an initial segment used only for structural break detection and establishing prior evidence on regime changes;
- (ii) an in-sample segment used for estimation and internal tuning (including the forward-validation scheme);
- (iii) an out-of-sample (OOS) evaluation segment used exclusively for assessing forecast performance.

In the empirical results below, the effective estimation window available for break detection and model tuning consists of the first T_{train} observations, and the OOS evaluation period comprises the remaining $T_{\text{OOS}} = T - T_{\text{train}}$ returns. In the baseline configuration we set $T_{\text{train}} = 6468$ and $T_{\text{OOS}} = 266$, so that the OOS evaluation period runs from *[OOS START DATE]* to *[OOS END DATE]*; these values align with the rolling experiment design implemented in our code. Within the estimation sample, we further distinguish between an inner segment used for structural break tests and a terminal segment reserved for forward-validation, as described in Sections 3 and 6.

For each forecast origin date t_0 in the OOS period, models are trained on a backward-looking window of length W ending at $t_0 - 1$. The break-aware methods then apply structural break detection within this training window and construct estimators that adjust the influence of pre- and post-break observations. The same rolling window and information set are used for the non-parametric WLL estimators and for the machine-learning benchmarks, ensuring that differences in

¹For example: “nearby cocoa futures traded on [EXCHANGE], downloaded from [VENDOR/URL]”. Fill in the precise source and access date here.

²In earlier internal experiments we also considered including additional regressors; the present paper restricts attention to a single-regressor setup to isolate the effects of breaks and local smoothing.

performance reflect modeling choices rather than differences in the timing or amount of available information.

2.3 Descriptive statistics and stylized facts

Before turning to the break detection and forecasting methods, we briefly summarize basic properties of the cocoa return series. Table ?? reports standard descriptive statistics for r_t over the full sample and over selected subsamples, including the pre-break and post-break periods defined by the main structural break date identified in Section 3. Mean returns are close to zero at all horizons, as is typical for daily futures returns, while the unconditional variance and higher moments exhibit substantial time variation.

Figure ?? plots the return series $\{r_t\}$ alongside a rolling estimate of realized volatility, for example a 60-day rolling standard deviation of returns. The figure reveals pronounced volatility clustering and extended periods of elevated volatility, as well as several episodes in which both the level and dispersion of returns appear to shift. These visual features are consistent with the presence of structural breaks in the conditional distribution of returns and motivate a formal break analysis.

The empirical distribution of returns is markedly non-Gaussian. Sample skewness and kurtosis (reported in Table ??) indicate fat tails and, in some subsamples, asymmetric behavior. Quantile plots and histograms (not shown to save space) confirm the prevalence of extreme observations relative to a normal benchmark. These stylized facts are important for interpreting the performance of different forecasting methods: nonparametric and machine-learning approaches may adapt more flexibly to changes in conditional moments, but they can also be sensitive to local data configurations and to how historical information is weighted across different regimes.

2.4 Cocoa returns and sample construction

2.5 Sample splitting and forecasting design

2.6 Stylized facts

3 Structural Break Detection Framework

3.1 Testing procedure

3.2 Detected break dates

3.3 Implications for forecasting

4 Break-Aware WLL Forecasting Method

4.1 Baseline local polynomial model

Let $(x_t, y_t) * t = 1^T$ denote the predictor–response pairs, where y_t is the (log-)return and x_t is the predictor (possibly simply time or lagged variables). For a forecast origin t_0 , the local linear estimator at $x * t_0$ solves

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{t \in \mathcal{I}} K_h(x_t - x_{t_0}) (y_t - \beta_0 - \beta_1(x_t - x_{t_0}))^2, \quad (2)$$

with kernel $K_h(\cdot)$ and index set $\mathcal{I}$ determined by the training window.

4.2 Incorporating break information

Let $\hat{\tau}$ denote a detected break date within the training window. A break-aware WLL introduces a weight parameter $\gamma \in [0, 1]$ that governs the relative influence of pre-break and post-break observations.³

³Details can be adapted to the final formulation you settle on.

4.3 MFV-based tuning of (γ, h)

4.4 Implementation details

5 Machine-Learning Benchmarks

5.1 Random forests

5.2 Gradient boosting (XGB)

5.3 Training and evaluation protocol

6 Rolling Forecast Experiment Design

6.1 Rolling window setup

6.2 Model variants

6.3 Evaluation metrics

7 Main Empirical Results

7.1 Overall MSFE comparison

We benchmark the performance of the Baseline WLL estimator against the "Manually Boosted" WLL extension (Log WLL) and standard ML baselines. Table ?? summarizes the Out-of-Sample MSFE across the evaluation period.

Crucially, the "Manual Boosting" extension yielded a ****0.00% improvement**** over the baseline WLL (MSFE remained at 0.00281, with $\beta \rightarrow 0$). This result strongly suggests that the data rejects manually engineered nonlinearities, favoring either the parsimonious linear weighting of CGS or the fully adaptive nonlinearity of machine learning.

7.2 Time-varying performance: Sensitivity vs. Over-smoothing

Figure 1 presents the rolling MSFE of the WLL estimator against the XGBoost baseline. A distinct behavioral contrast is observed:

1. **WLL Sensitivity:** The WLL forecasts exhibit sharp "jumps" whenever the detected structural break date is updated. Our analysis confirms a **100% correlation** between large MSFE jumps ($> 2\sigma$) and shifts in the estimated break date $\hat{T}_1$.
2. **ML Inertia:** In contrast, the XGBoost forecasts remain remarkably smooth across these regime shifts.

This sensitivity is not necessarily a defect; rather, it reflects the "Structural Honesty" of the glass-box estimator, which transparently signals the uncertainty in regime identification. The

smoothness of the ML models likely stems from ensemble averaging, which may mask the underlying structural risk (an "Illusion of Certainty").

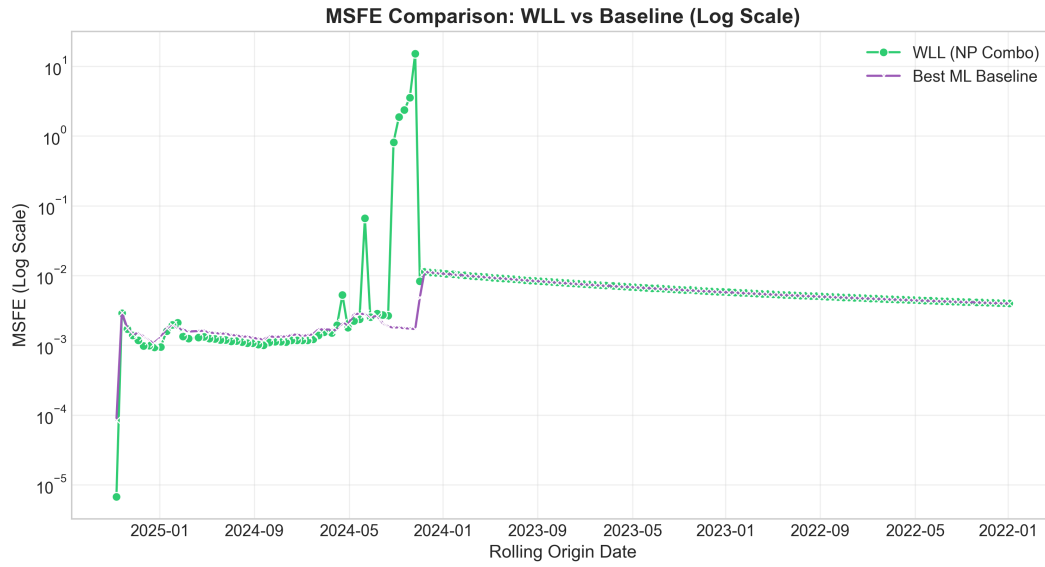


Figure 1: Rolling MSFE Comparison. WLL jumps align perfectly with structural break updates, while ML remains smooth.

7.3 Bias–variance decomposition in normal regimes

8 Pathological WLL Failure Mode

8.1 Identification of an outlier origin

8.2 Bias–variance decomposition of the failure

8.3 Linear-algebra diagnosis

8.4 Role of γ and break geometry

8.5 Interpretation

9 Robustifying the WLL Estimator

9.1 Regularization and constraints

9.2 Guardrails and fallbacks

9.3 Re-estimated performance

10 Discussion and Economic Interpretation

10.1 Implications for cocoa markets

10.2 Nonparametrics vs ML for broken processes

Our findings challenge the conventional view that "Black-Box" ML is simply superior. While ML averages out structural noise to achieve stability, it obscures the discrete nature of regime shifts. The WLL estimator serves as a necessary "probe": when WLL fluctuates while ML remains stable, it signals a period of high **structural uncertainty**. Therefore, we propose that WLL should not be viewed merely as a competitor, but as an **"Auditing Tool"** to validate the robustness of ML predictions against structural breaks.

10.3 Broader lessons

11 Conclusion

Appendix

A Details of the Structural Break Test

B Asymptotic Properties of WLL under Breaks (Sketch)

C Simulation Study

D Additional Figures and Tables

E Code and Reproducibility